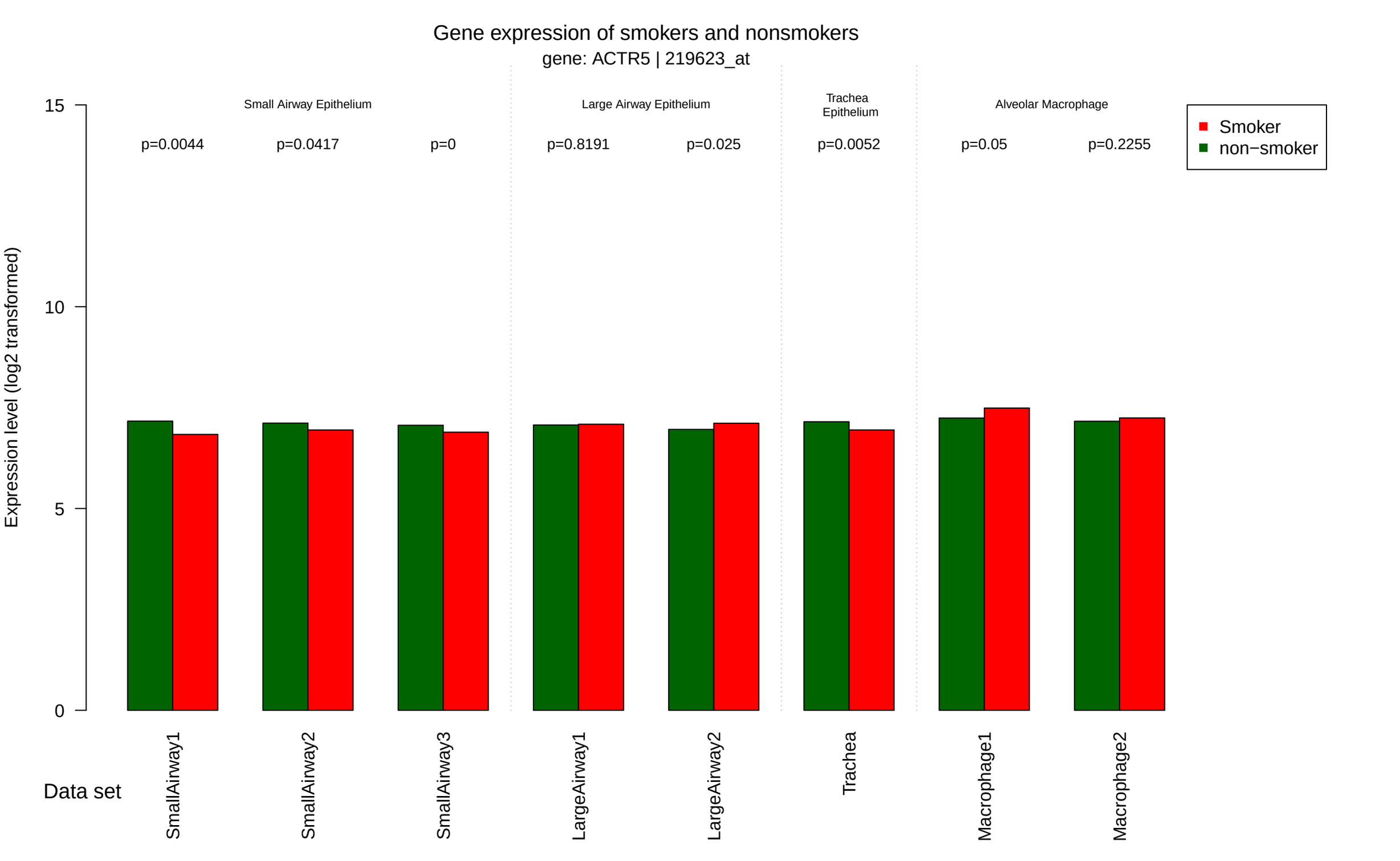
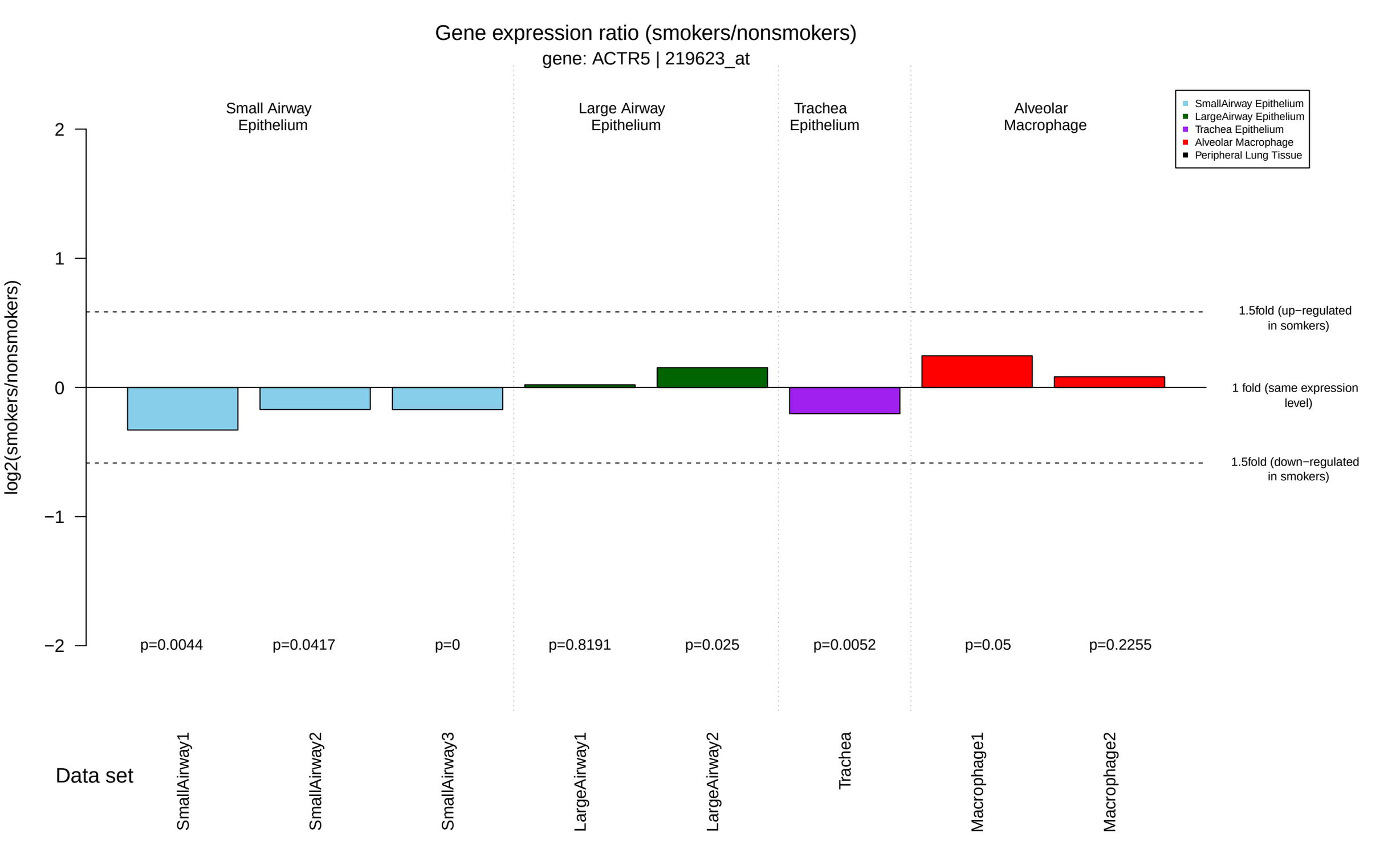
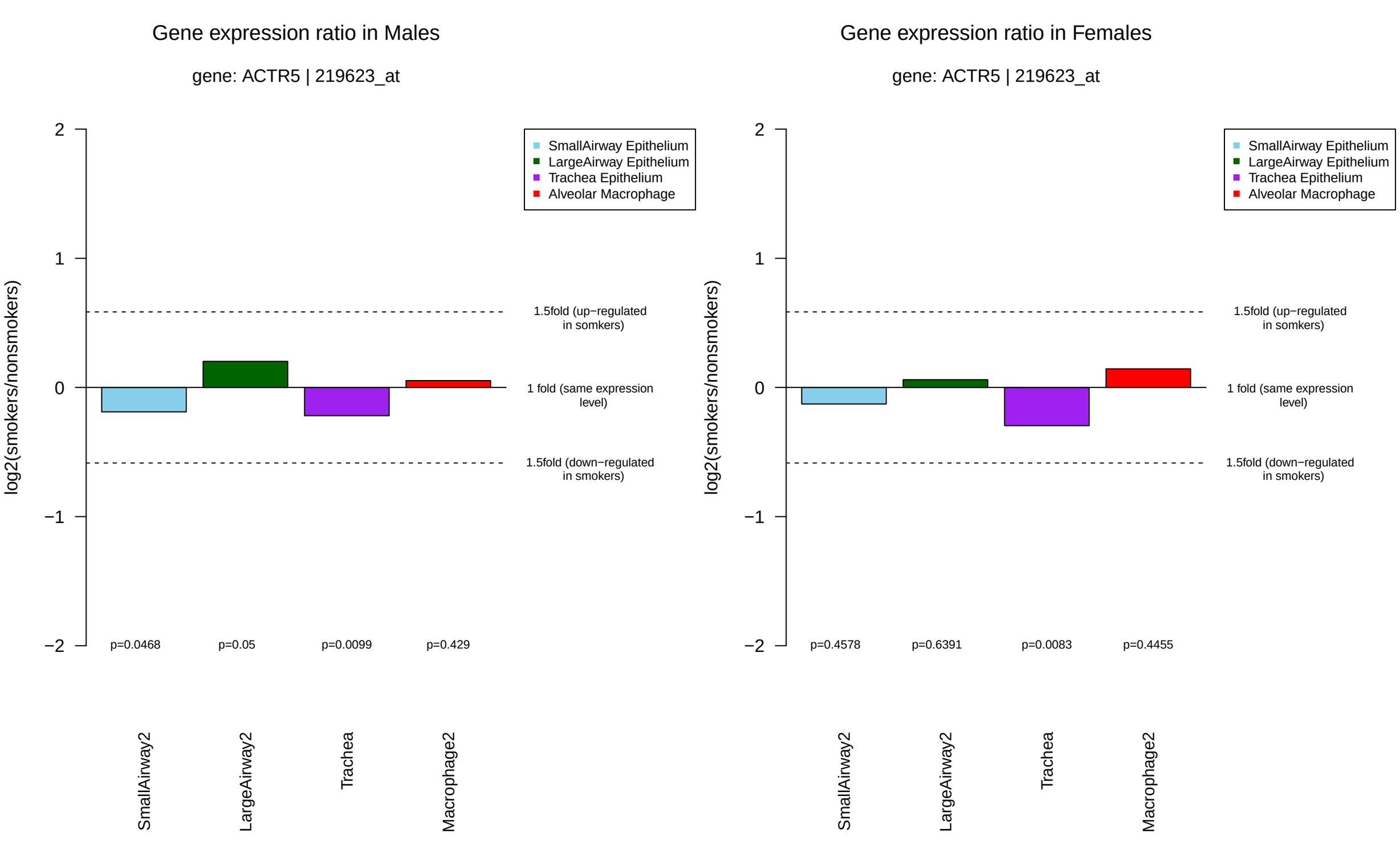
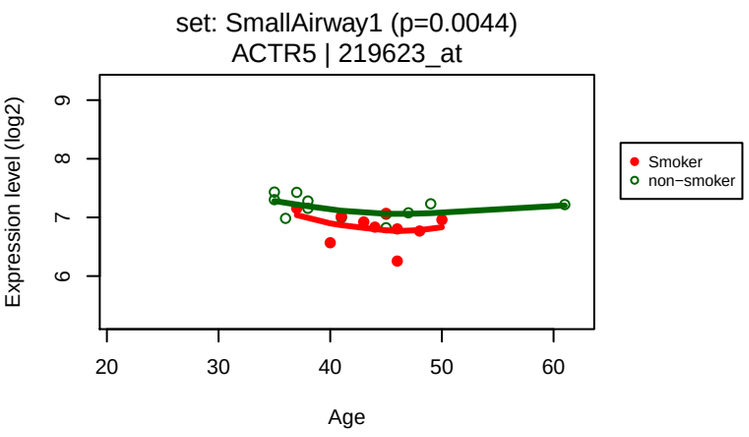
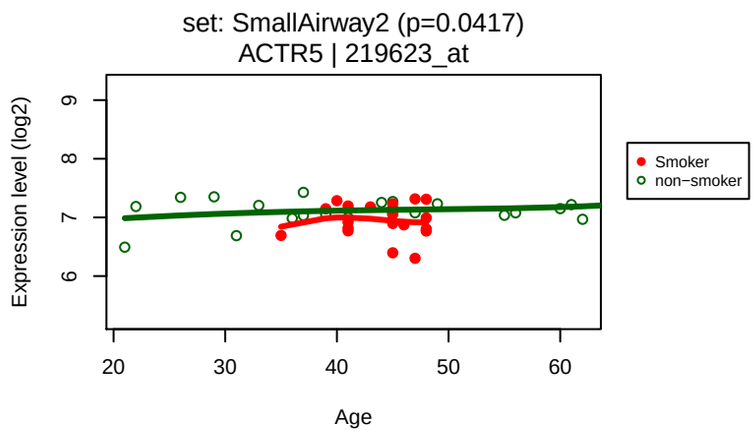
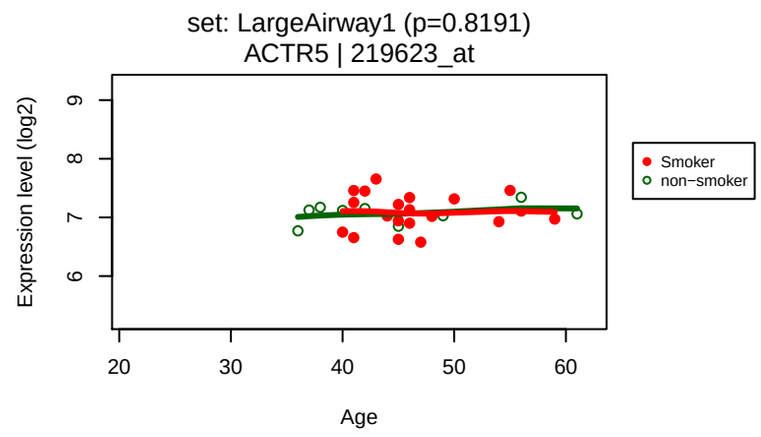

Gene expression of smokers and nonsmokers
gene: ACTR5 | 219623_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0044
p=0.0417
p=0
p=0.8191
p=0.025
p=0.0052
p=0.05
p=0.2255
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

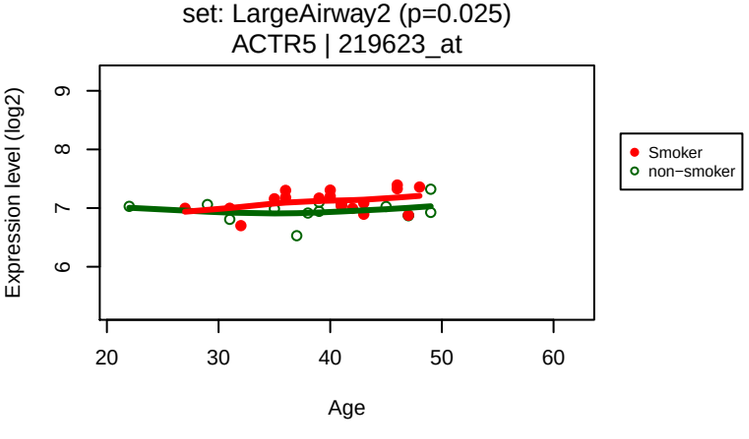

Gene expression ratio (smokers/nonsmokers)
gene: ACTR5 | 219623_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0044
p=0.0417
p=0
p=0.8191
p=0.025
p=0.0052
p=0.05
p=0.2255
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

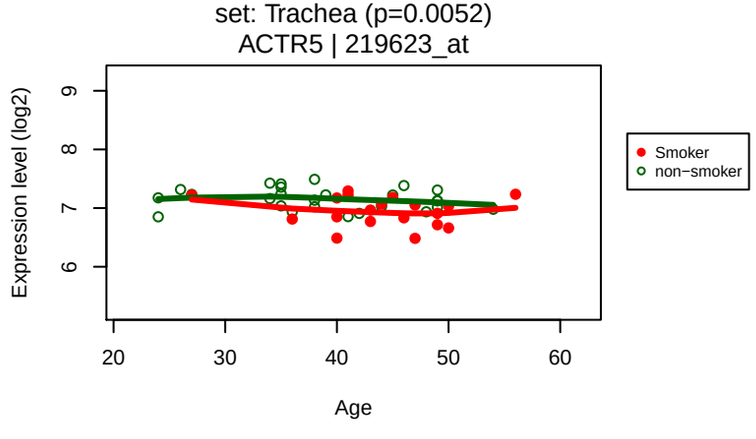

Gene expression ratio in Males
gene: ACTR5 | 219623_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0468
p=0.05
p=0.0099
p=0.429
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR5 | 219623_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4578
p=0.6391
p=0.0083
p=0.4455
SmallAirway2
LargeAirway2
Trachea
Macrophage2

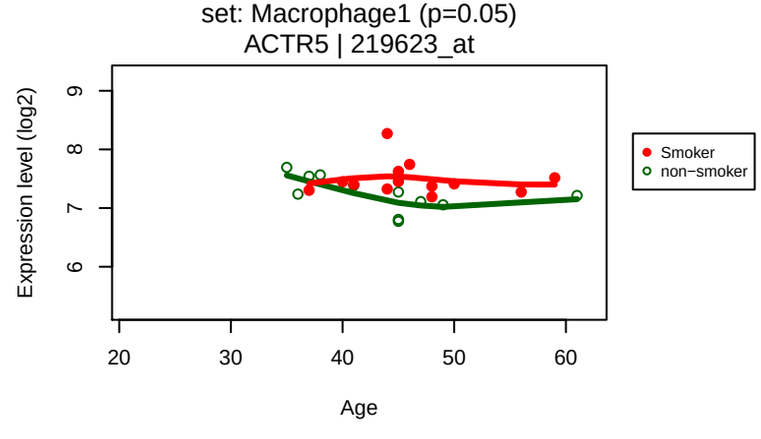

set: SmallAirway1 (p=0.0044)
ACTR5 | 219623_at
Expression level (log2)
Age
Smoker
non−smoker

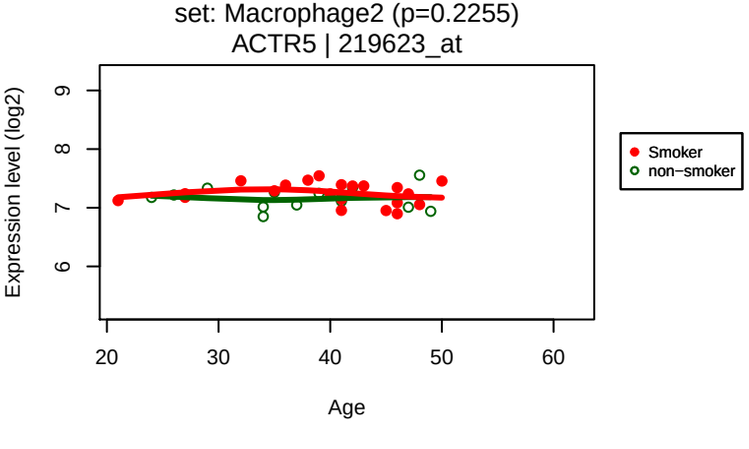

set: SmallAirway2 (p=0.0417)
ACTR5 | 219623_at
Expression level (log2)
Age
Smoker
non−smoker

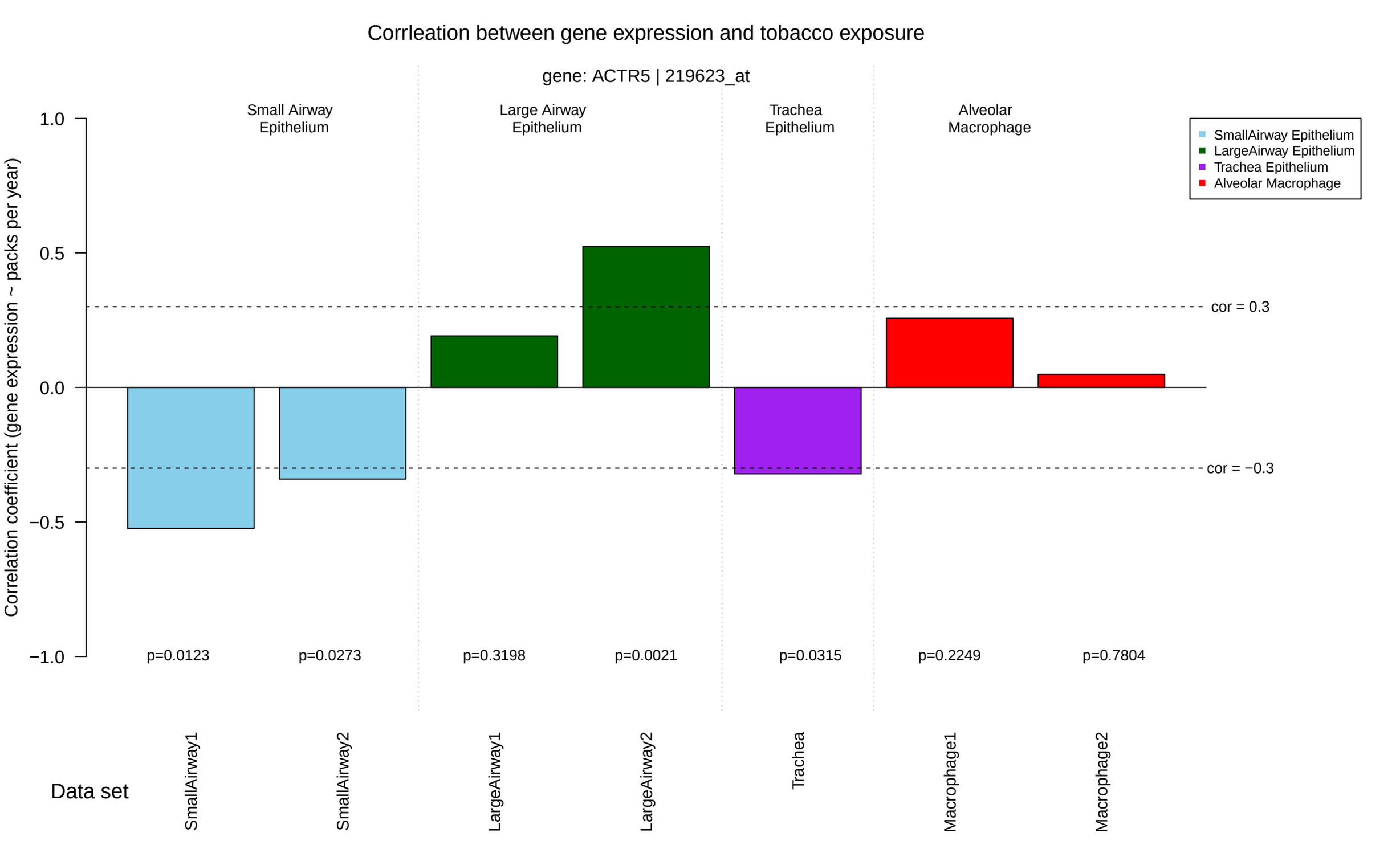

set: LargeAirway1 (p=0.8191)
ACTR5 | 219623_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.025)
ACTR5 | 219623_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0052)
ACTR5 | 219623_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.05)
ACTR5 | 219623_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2255)
ACTR5 | 219623_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR5 | 219623_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0123
p=0.0273
p=0.3198
p=0.0021
p=0.0315
p=0.2249
p=0.7804
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2